

FDU 回归分析 期中考试 (2024 秋)

Total: 100 marks

Duration: 2 hour and 30 minutes

Problem 1

Let Y_1, Y_2, Y_3 be independent response observations satisfying:

$$Y_i = \begin{cases} \beta + \varepsilon_i & \text{if } i = 1, 2 \\ -\beta + \varepsilon_i & \text{if } i = 3 \end{cases}$$

where $\beta \in \mathbb{R}$ is an unknown parameter and $\varepsilon_1, \varepsilon_2, \varepsilon_3$ are independent $N(0, \sigma^2)$ variables for some unknown $\sigma^2 > 0$

Part (1)

The above setting describes a linear model of the form:

$$Y \equiv \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} = X\beta + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}$$

where $X \in \mathbb{R}^{3 \times 1}$ denotes the design matrix.

Find X . [2 Marks]

Solution:

$$X = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Part (2)

Show that the least squares estimator of β is $\hat{\beta} = \frac{1}{3}(Y_1 + Y_2 - Y_3)$ [4 Marks]

Solution:

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T Y \\ &= \left(\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}^T \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} \\ &= \frac{1}{3}(Y_1 + Y_2 - Y_3) \end{aligned}$$

Part (3)

The fitted values of Y_1, Y_2, Y_3 are given by the vector $\hat{Y} = HY$ for some matrix $H \in \mathbb{R}^{3 \times 3}$.

Find H . [2 Marks]

Solution:

$$\begin{aligned}
H &= X(X^T X)^{-1} X^T \\
&= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \left(\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}^T \\
&= \frac{1}{3} \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}
\end{aligned}$$

Part (4)

Show that the residual sum of squares has a quadratic form $Y^T A Y$ for some $A \in \mathbb{R}^{3 \times 3}$.
Find A . [4 Marks]

Solution:

注意到:

$$\begin{aligned}
\text{SSE} &= \|Y - \hat{Y}\|^2 \\
&= \|Y - HY\|^2 \\
&= Y^T (I - H)^T (I - H) Y \quad (\text{note that } \begin{cases} H^T = H \\ H^2 = H \end{cases}) \\
&= Y^T (I - H) Y
\end{aligned}$$

因此我们有:

$$\begin{aligned}
A &= I - H \\
&= \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \\
&= \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}
\end{aligned}$$

Part (5)

Show that the residual sum of squares and $\hat{Y}$ are independent. [4 Marks]

Solution:

注意到:

$$\begin{aligned}
\text{SSE} &= \|Y - \hat{Y}\|^2 \\
&= \|Y - HY\|^2 \\
&= Y^T (I - H)^T (I - H) Y \\
&= Y^T (I - H) Y \\
&= (X\beta + \varepsilon)^T (I - H) (X\beta + \varepsilon) \quad (\text{note that } HX = X \Rightarrow (I - H)X = 0_{n \times (p+1)}) \\
&= \varepsilon^T (I - H) \varepsilon \\
&= \|(I - H)\varepsilon\|^2
\end{aligned}$$

而 $\hat{Y} = HY = H(X\beta + \varepsilon) = X\beta + H\varepsilon$

要证明 $\text{SSE} \perp \hat{Y}$, 只要证明 $(I - H)\varepsilon \perp H\varepsilon$

由于 $(I - H)\varepsilon$ 和 $H\varepsilon$ 都是 $\varepsilon \sim N(0_n, \sigma^2 I_n)$, 故它们联合正态.

因此只要证明 $(I - H)\varepsilon$ 和 $H\varepsilon$ 不相关即可:

$$\begin{aligned}
\text{Cov}((I - H)\varepsilon, H\varepsilon) &= (I - H)\text{Cov}(\varepsilon)H \\
&= (I - H) \cdot \sigma^2 I_n \cdot H \\
&= \sigma^2(H - H^2) \\
&= \sigma^2(H - H) \\
&= 0_{n \times n}
\end{aligned}$$

因此 $\text{SSE} \perp \hat{Y}$

Part (6)

Based on the result in part (5),

test whether or not $\beta = 0$ at the 5% level of significance given that $\begin{cases} Y_1 = 8 \\ Y_2 = 12 \\ Y_3 = -13 \end{cases}$ [4 Marks]

Solution:

注意到:

$$\begin{aligned}
\hat{\beta} &= (X^T X)^{-1} X^T Y \\
&= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\
&= \beta + (X^T X)^{-1} X^T \varepsilon \\
&= \beta + N((X^T X)^{-1} X^T 0_n, (X^T X)^{-1} X^T \cdot \sigma^2 I_n \cdot X (X^T X)^{-1}) \Rightarrow \frac{\hat{\beta} - \beta}{\sigma \sqrt{(X^T X)^{-1}}} \sim N(0, 1) \\
&= \beta + N(0_n, \sigma^2 (X^T X)^{-1}) \\
&= N(\beta, \sigma^2 (X^T X)^{-1})
\end{aligned}$$

代入 $X = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ 和 $Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \\ -13 \end{bmatrix}$ 可知:

$$\hat{\beta} = (X^T X)^{-1} X^T Y = \frac{1}{3} (Y_1 + Y_2 - Y_3) = \frac{1}{3} (8 + 12 - (-13)) = 11$$

$$s^2 = \frac{1}{2} \text{SSE} = \frac{1}{2} Y^T A Y = \frac{1}{2} \begin{bmatrix} 8 \\ 12 \\ -13 \end{bmatrix}^T \left(\frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \right) \begin{bmatrix} 8 \\ 12 \\ -13 \end{bmatrix} = 7$$

注意到 $\text{tr}(A) = \text{tr} \left(\frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \right) = 2$, 因此我们有 $s^2 \sim \sigma^2 \chi_{(2)}^2$

因此我们有:

$$\frac{\hat{\beta} - \beta}{s \sqrt{(X^T X)^{-1}}} = \frac{\frac{\hat{\beta} - \beta}{\sigma \sqrt{(X^T X)^{-1}}}}{\frac{s}{\sigma}} \sim \frac{N(0, 1)}{\sqrt{\chi_{(2)}^2/2}} = t_{(2)}$$

在零假设 $H_0: \beta = 0$ 下我们有:

$$\left| \frac{\hat{\beta} - \beta}{s \sqrt{(X^T X)^{-1}}} \right| \stackrel{H_0}{=} \left| \frac{11 - 0}{\sqrt{7} \cdot \sqrt{\frac{1}{3}}} \right| = \left| 11 \sqrt{\frac{3}{7}} \right| \approx 7.20$$

注意到 $F_{1,2}(0.05) \approx 18.51$, 因此 $t_2(0.025) = \sqrt{F_{1,2}(0.05)} = \sqrt{18.51} \approx 4.30$

于是我们有:

$$\left| \frac{\hat{\beta} - \beta}{s\sqrt{(X^T X)^{-1}}} \right| \stackrel{H_0}{=} \left| \frac{11 - 0}{\sqrt{7} \cdot \sqrt{\frac{1}{3}}} \right| = \left| 11\sqrt{\frac{3}{7}} \right| \approx 7.20 > 4.30 = t_2(0.025)$$

这表明在因子水平 $\alpha = 0.05$ 下我们可以拒绝零假设 $H_0 : \beta = 0$

Problem 2

Consider the linear regression model:

$$Y \equiv \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \sim N(X\beta, \sigma^2 I_n)$$

$$Y_* \equiv \begin{bmatrix} Y_1^* \\ \vdots \\ Y_m^* \end{bmatrix} \sim N(X_*\beta, \sigma^2 I_m)$$

where $X \in \mathbb{R}^{n \times d}$, $X_* \in \mathbb{R}^{m \times d}$ are given design matrices, and observation $Y \in \mathbb{R}^n$ is given.

Let $a = [a_1, \dots, a_m]^T$ be a vector of m known constants.

Suppose we are interested in predicting $l = a^T Y_* = \sum_{j=1}^m a_j Y_j^*$

Give the $1 - \alpha$ prediction interval for l [10 Marks]

Solution:

模型的最小二乘估计量为 $\hat{\beta} := (X^T X)^{-1} X^T Y \sim N(\beta, \sigma^2 (X^T X)^{-1})$

定义 $\hat{l} := a^T \hat{Y}_* = a^T X_* \hat{\beta} = a^T X_* (X^T X)^{-1} X^T Y$

则我们有:

$$\begin{aligned} \hat{l} - l &= a^T X_* \hat{\beta} - a^T Y_* \\ &= a^T X_* \hat{\beta} - a^T (X_* \beta + \varepsilon_*) \\ &= a^T X_* (\hat{\beta} - \beta) - a^T \varepsilon_* \quad \left(\text{note that } \begin{cases} \hat{\beta} - \beta \sim N(0_d, \sigma^2 (X^T X)^{-1}) \\ \varepsilon_* \sim N(0_m, \sigma^2 I_m) \\ \hat{\beta} \perp \varepsilon_* \end{cases} \right) \\ &\sim N(a^T X_* 0_d - a^T 0_m, a^T X_* \cdot \sigma^2 (X^T X)^{-1} \cdot X_*^T a + a^T \cdot \sigma^2 I_m \cdot a) \\ &= N(0, \sigma^2 a^T [X_* (X^T X)^{-1} X_*^T + I_m] a) \end{aligned}$$

其中 $\hat{\beta} \perp \varepsilon_*$ 是因为 $\hat{\beta}$ 依赖于前 n 个样本, 与新的 m 个样本的随机噪音 ε_* 相互独立.

因此我们有:

$$\frac{l - \hat{l}}{\sigma \sqrt{a^T [X_* (X^T X)^{-1} X_*^T + I_m] a}} \sim N(0, 1)$$

现考虑 $s^2 = \frac{1}{n-d} \|Y - \hat{Y}\|^2 = \frac{1}{n-d} \|Y - HY\|^2 = \frac{1}{n-d} Y^T (I_n - H) Y$ (其中 $H = X(X^T X)^{-1} X^T$)

我们有 $\begin{cases} s^2 \sim \frac{1}{n-d} \sigma^2 \chi_{(n-d)}^2 \\ s^2 \perp \hat{\beta} \\ s^2 \perp \varepsilon_* \end{cases}$ (其中 $s^2 \perp \varepsilon_*$ 是因为 s^2 依赖于前 n 个样本, 与新的 m 个样本的随机噪音 ε_* 相互独立)

于是我们有:

$$\frac{l - \hat{l}}{s \sqrt{a^T [X_* (X^T X)^{-1} X_*^T + I_m] a}} = \frac{\frac{l - \hat{l}}{\sigma \sqrt{a^T [X_* (X^T X)^{-1} X_*^T + I_m] a}}}{\frac{s}{\sigma}} \sim \frac{N(0, 1)}{\sqrt{\chi_{(n-d)}^2 / (n-d)}} = t_{(n-d)}$$

因此 l 的 $100(1 - \alpha)\%$ 预测区间为 $[\hat{l} \pm t_{n-d, \frac{\alpha}{2}} \cdot s \sqrt{a^T [X_\star (X^T X)^{-1} X_\star^T + I_m] a}]$

其中 $\hat{l} := a^T \hat{Y}_\star = a^T X_\star \hat{\beta} = a^T X_\star (X^T X)^{-1} X^T Y$

而 $t_{n-d, \frac{\alpha}{2}}$ 是 $t_{(n-d)}$ 分布的 $1 - \frac{\alpha}{2}$ 分位数.

Problem 3

Suppose that $\mu \equiv \mathbb{E}[Y] = X\beta$ and $\text{Cov}(Y) = \sigma^2 I_n$ in the Linear regression model, where $X \in \mathbb{R}^{n \times (p+1)}$ and $\beta \in \mathbb{R}^{p+1}$

Let $\tilde{\phi} = c^T Y$ be any unbiased linear estimator of $\phi = t^T \beta$, where $t \in \mathbb{R}^{p+1}$ is an arbitrary vector.

Prove that $\text{Var}(\tilde{\phi}) \geq \text{Var}(\hat{\phi})$

where $\hat{\phi} = t^T \hat{\beta}$ and $\hat{\beta} = (X^T X)^{-1} X^T y$ is the least squares estimator of β . [10 Marks]

Solution:

根据 $\tilde{\phi} = c^T Y$ 的无偏性可知:

$$\mathbb{E}[\tilde{\phi}] = \mathbb{E}[c^T Y] = c^T \mathbb{E}[Y] = c^T (X\beta) = t^T \beta$$

因此 $c \in \mathbb{R}^n$ 一定满足 $X^T c = t$

目标函数 $\text{Var}(\tilde{\phi}) = \text{Var}(c^T Y) = c^T \text{Var}(Y) c = c^T \cdot \sigma^2 I_n \cdot c = \sigma^2 c^T c$

考虑求解最优化问题:

$$\min_{X^T c = t} c^T c$$

定义 Lagrange 函数为 $L(c, \lambda) = c^T c - \lambda^T (X^T c - t)$ (其中 Lagrange 乘子 $\lambda \in \mathbb{R}^{p+1}$)

注意到上述问题是凸优化问题, 故其最优解即 KKT 点.

其 KKT 条件为:

$$\begin{cases} \nabla_c L(c, \lambda) = 2c - X\lambda = 0_n \\ X^T c = t \end{cases}$$

将 $c = \frac{1}{2} X\lambda$ 代入 $X^T c = t$ 即得 $\lambda_\star = 2(X^T X)^{-1} t$

因此 $c_\star = \frac{1}{2} X\lambda_\star = X(X^T X)^{-1} t$

这意味着使得 $\text{Var}(\tilde{\phi}) = \sigma^2 c^T c$ 达到最小值的 $\tilde{\phi}_\star = c_\star^T Y = t^T (X^T X)^{-1} X^T Y = t^T \hat{\beta} = \hat{\phi}$

所以我们有 $\text{Var}(\tilde{\phi}) \geq \text{Var}(\hat{\phi})$ 恒成立.

Problem 4

Consider a multiple linear regression model:

$$Y = \beta_0 1_n + X\beta + \varepsilon \text{ where } \begin{cases} \mathbb{E}[\varepsilon] = 0_n \\ \text{Cov}[\varepsilon] = \sigma^2 I_n \end{cases}$$

where $Y \in \mathbb{R}^n$ is the observation vector and $X \in \mathbb{R}^{n \times k}$ design matrix with full rank.

Find the F-statistics for the following two hypothesis testing problems:

- ① $H_0 : \beta_1 = \cdots = \beta_k = c$ where $c \in \mathbb{R}$ is some given constant. [10 Marks]
- ② $H_0 : \beta_1 = \cdots = \beta_k$ [10 Marks]

Part (1)

$H_0 : \beta_1 = \cdots = \beta_k = c$ where $c \in \mathbb{R}$ is some given constant. [10 Marks]

Solution:

零假设 $H_0 : \beta_1 = \cdots = \beta_k = c$ 可以等价写为:

$$[0_k, I_k] \begin{bmatrix} \beta_0 \\ \beta \end{bmatrix} = c \mathbf{1}_k$$

表明等式约束的个数为 k

在 H_0 成立条件下的简约模型为 $Y = \beta_0 \mathbf{1}_n + X \cdot c \mathbf{1}_k + \varepsilon$

我们记 $\tilde{Y} := Y - X \cdot c \mathbf{1}_k$ 即得 $\tilde{Y} = \beta_0 \mathbf{1}_n + \varepsilon$

因此关于简约模型我们有:

$$\hat{\beta}_0 := (\mathbf{1}_n^T \mathbf{1}_n)^{-1} \mathbf{1}_n^T \tilde{Y} = \frac{1}{n} \mathbf{1}_n^T \tilde{Y} \text{ where } \tilde{Y} = Y - X \cdot c \mathbf{1}_k$$
$$\text{SSE}_{\text{reduced}} := \|\tilde{Y} - \hat{\beta}_0 \mathbf{1}_n\|^2$$

而关于全模型我们有:

$$\tilde{X} := [\mathbf{1}_n, X] \in \mathbb{R}^{n \times (k+1)}$$
$$\hat{\beta}_{\text{full}} := (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T Y$$
$$\text{SSE}_{\text{full}} := \|Y - \tilde{X} \hat{\beta}_{\text{full}}\|^2$$

额外平方和 $\text{ESS} := \text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}} \sim \sigma^2 \chi_{(k)}^2$ (考虑到等式约束的个数为 k 所以其自由度为 k)

因此我们可以定义 F-统计量:

$$F := \frac{\text{ESS}/k}{\text{SSE}_{\text{full}}/(n-k-1)} \sim F_{k, n-k-1}$$

检验法为:

若 $F > F_{k, n-k-1}(\alpha)$, 就拒绝零假设 $H_0 : \beta_1 = \cdots = \beta_k = c$

其中 $F_{k, n-k-1}(\alpha)$ 为 $F_{k, n-k-1}$ 分布的 $1 - \alpha$ 分位数.

Part (2)

$H_0 : \beta_1 = \cdots = \beta_k$ [10 Marks]

Solution:

零假设 $H_0 : \beta_1 = \cdots = \beta_k$ 可以等价写为:

$$C \begin{bmatrix} \beta_0 \\ \beta \end{bmatrix} = \begin{bmatrix} 0 & 1 & -1 & & & \\ 0 & & 1 & -1 & & \\ \vdots & & & \ddots & \ddots & \\ 0 & & & & 1 & -1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

其中系数矩阵 $C \in \mathbb{R}^{(k-1) \times (k+1)}$, 表明等式约束的个数为 $k-1$

在 H_0 成立条件下的简约模型为 $Y = \beta_0 \mathbf{1}_n + X \mathbf{1}_k \beta_1 + \varepsilon$

我们记 $X_1 := [\mathbf{1}_n, X \mathbf{1}_k] \in \mathbb{R}^{n \times 2}$ 即得 $Y = X_1 \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \varepsilon$

因此关于简约模型我们有:

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} := (X_1^T X_1)^{-1} X_1^T Y \text{ where } X_1 := [1_n, X1_k]$$

$$\text{SSE}_{\text{reduced}} := \left\| Y - X_1 \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} \right\|^2$$

而关于全模型我们有:

$$\tilde{X} := [1_n, X] \in \mathbb{R}^{n \times (k+1)}$$

$$\hat{\beta}_{\text{full}} := (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T Y$$

$$\text{SSE}_{\text{full}} := \|Y - \tilde{X} \hat{\beta}_{\text{full}}\|^2$$

额外平方和 $\text{ESS} := \text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}} \sim \sigma^2 \chi_{(k-1)}^2$ (考虑到等式约束的个数为 $k-1$ 所以其自由度为 $k-1$)

(实际上根据 Problem 5 的结论我们可以写出确切的表达式: $\text{ESS} = (C \hat{\beta}_{\text{full}})^T [C(X^T X)^{-1} C^T]^{-1} (C \hat{\beta}_{\text{full}})$)

因此我们可以定义 F-统计量:

$$F := \frac{\text{ESS}/(k-1)}{\text{SSE}_{\text{full}}/(n-k-1)} \sim F_{k-1, n-k-1}$$

检验法为:

若 $F > F_{k-1, n-k-1}(\alpha)$, 就拒绝零假设 $H_0: \beta_1 = \dots = \beta_k$

其中 $F_{k-1, n-k-1}(\alpha)$ 为 $F_{k-1, n-k-1}$ 分布的 $1-\alpha$ 分位数.

Problem 5

Given the multiple linear regression model with k independent variables:

$$y = X\beta + \varepsilon \text{ where } \varepsilon \sim N(0, \sigma^2 I_n)$$

Consider the following hypothesis testing:

$$H_0: C\beta = h \quad \Leftrightarrow \quad H_1: C\beta \neq h$$

where $C \in \mathbb{R}^{m \times (k+1)}$ is a rank- m constant matrix and $h \in \mathbb{R}^m$ is a constant vector.

(1) Find the LSE (least squares estimator) of β under H_0 [10 marks]

(2) Find the F-test for testing H_0

(Write the test statistic and prove its null distribution) [10 marks]

Part (1)

Find the LSE (least squares estimator) of β under H_0 [10 marks]

Solution:

考虑求解线性约束最小二乘问题 $\min_{C\beta=h} \|y - X\beta\|_2^2$

注意到目标函数 $f(\beta) = \|y - X\beta\|_2^2$ 是关于 β 的凸函数, 而问题只有线性等式约束 $C\beta = h$ 因此这是一个标准形式的凸优化问题, 其最优解即为 KKT 点.

定义其 Lagrange 函数 $L(\beta, \lambda)$ 为:

$$\begin{aligned} L(\beta, \lambda) &= f(\beta) - \lambda^T (C\beta - h) \\ &= \|y - X\beta\|_2^2 - \lambda^T (C\beta - h) \end{aligned}$$

$$\text{dom}\{L\} = \mathbb{R}^{p+1} \times \mathbb{R}^m$$

Lagrange 函数 $L(\beta, \lambda)$ 关于 β 的梯度为:

$$\begin{aligned}
\nabla_{\beta} L(\beta, \lambda) &= \nabla_{\beta} \{ \|y - X\beta\|_2^2 - \lambda^T (C\beta - h) \} \\
&= -X^T \cdot 2(y - X\beta) - (\lambda^T C)^T \\
&= -2X^T y + 2X^T X\beta - C^T \lambda
\end{aligned}$$

KKT 条件为:

$$\begin{cases} \nabla_{\beta} L(\beta, \lambda) = -2X^T y + 2X^T X\beta - C^T \lambda = 0_{p+1} & \text{①} \\ C\beta = h & \text{②} \end{cases}$$

① 式左乘 $(X^T X)^{-1}$ 可得 $-2(X^T X)^{-1} X^T y + 2\beta - (X^T X)^{-1} C^T \lambda = 0_{p+1}$

于是有 $\beta = (X^T X)^{-1} X^T y + \frac{1}{2} (X^T X)^{-1} C^T \lambda$

代入 ② 式即得 $C\beta = C(X^T X)^{-1} X^T y + \frac{1}{2} C(X^T X)^{-1} C^T \lambda = h$

解得 $\lambda_{\text{KKT}} = 2[C(X^T X)^{-1} C^T]^{-1} [h - C(X^T X)^{-1} X^T y]$

因此我们有:

$$\begin{aligned}
\hat{\beta}_{\text{reduced}} &= \beta_{\text{KKT}} \\
&= (X^T X)^{-1} X^T y + \frac{1}{2} (X^T X)^{-1} C^T \lambda_{\text{KKT}} \\
&= (X^T X)^{-1} X^T y + \frac{1}{2} (X^T X)^{-1} C^T \cdot 2[C(X^T X)^{-1} C^T]^{-1} [h - C(X^T X)^{-1} X^T y] \\
&= \hat{\beta}_{\text{full}} - (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h)
\end{aligned}$$

其中 $\hat{\beta}_{\text{full}} = (X^T X)^{-1} X^T y$

Part (2)

Find the F-test for testing H_0

(Write the test statistic and prove its null distribution) [10 marks]

Solution:

下面我们计算简约模型误差平方和 $\text{SSE}_{\text{reduced}} = \|y - X\hat{\beta}_{\text{reduced}}\|_2^2$

$$\begin{aligned}
\text{SSE}_{\text{reduced}} &= \|y - X\hat{\beta}_{\text{reduced}}\|_2^2 \\
&= \|y - X(\hat{\beta}_{\text{full}} - (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h))\|_2^2 \\
&= \|y - X\hat{\beta}_{\text{full}} + X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h)\|_2^2 \\
&= \|y - X\hat{\beta}_{\text{full}} + A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \quad (\text{denote } A := X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}) \\
&= \|y - X\hat{\beta}_{\text{full}}\|_2^2 - 2(y - X\hat{\beta}_{\text{full}})^T A(C\hat{\beta}_{\text{full}} - h) + \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2
\end{aligned}$$

其中 $A := X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} \in \mathbb{R}^{n \times m}$.

考虑交叉项 $(y - X\hat{\beta}_{\text{full}})^T A(C\hat{\beta}_{\text{full}} - h)$:

$$\begin{aligned}
&(y - X\hat{\beta}_{\text{full}})^T A(C\hat{\beta}_{\text{full}} - h) \\
&= (y - Hy)^T A(C\hat{\beta}_{\text{full}} - h) \quad (\text{recall that } X\hat{\beta}_{\text{full}} = Hy \text{ where } H = X(X^T X)^{-1} X^T y) \\
&= y^T (I_n - H) X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h) \quad (\text{note that } HX = X \text{ so that } (I_n - H)X = 0_{n \times (p+1)}) \\
&= y^T 0_{n \times (p+1)} (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h) \\
&= 0
\end{aligned}$$

因此我们有:

$$\begin{aligned}
\text{SSE}_{\text{reduced}} &= \|y - X\hat{\beta}_{\text{full}}\|_2^2 - 2(y - X\hat{\beta}_{\text{full}})^T A(C\hat{\beta}_{\text{full}} - h) + \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \\
&= \text{SSE}_{\text{full}} - 2 \cdot 0 + \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \\
&= \text{SSE}_{\text{full}} + \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2
\end{aligned}$$

我们定义**额外误差平方和** (extra sum of squares, ESS) 为从全模型到简约模型增加的误差平方和:

$$\text{ESS} = \text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}} = \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2$$

其中 $\begin{cases} A = X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} \\ \hat{\beta}_{\text{full}} = (X^T X)^{-1} X^T y \end{cases}$

考虑第一类错误概率界限为 α 的检验问题 $H_0 : C\beta = h \leftrightarrow H_1 : C\beta \neq h$

下面我们研究**额外误差平方和** $\text{ESS} = \text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}} = \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2$ 在零假设 $H_0 : C\beta = h$ 下的分布.

$$\begin{aligned} \text{ESS} &= \text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}} \\ &= \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \\ &= (C\hat{\beta}_{\text{full}} - h)^T A^T A (C\hat{\beta}_{\text{full}} - h) \quad (\text{recall that } A = X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}) \\ &= (C\hat{\beta}_{\text{full}} - h)^T \{X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}\}^T \{X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}\} (C\hat{\beta}_{\text{full}} - h) \\ &= (C\hat{\beta}_{\text{full}} - h)^T \{[C(X^T X)^{-1} C^T]^{-1} C(X^T X)^{-1} X^T\} \cdot \{X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}\} (C\hat{\beta}_{\text{full}} - h) \\ &= (C\hat{\beta}_{\text{full}} - h)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h) \\ &= \eta^T \eta \quad (\text{denote } \eta = [C(X^T X)^{-1} C^T]^{-\frac{1}{2}} (C\hat{\beta}_{\text{full}} - h)) \end{aligned}$$

注意到 $\hat{\beta}_{\text{full}} = (X^T X)^{-1} X^T y \sim N(\beta, \sigma^2 (X^T X)^{-1})$

于是我们有:

$$\begin{aligned} C\hat{\beta}_{\text{full}} - h &\sim N(C\beta - h, \sigma^2 C(X^T X)^{-1} C^T) \\ &\stackrel{H_0}{=} N(0_m, \sigma^2 C(X^T X)^{-1} C^T) \quad (\text{where } H_0 : C\beta = h) \end{aligned}$$

因此我们有:

$$\begin{aligned} \eta &= [C(X^T X)^{-1} C^T]^{-\frac{1}{2}} (C\hat{\beta}_{\text{full}} - h) \\ &\stackrel{H_0}{\sim} N([C(X^T X)^{-1} C^T]^{-\frac{1}{2}} \cdot 0_m, [C(X^T X)^{-1} C^T]^{-\frac{1}{2}} \sigma^2 C(X^T X)^{-1} C^T \{[C(X^T X)^{-1} C^T]^{-\frac{1}{2}}\}^T) \quad (\text{where } H_0 : C\beta = h) \\ &= N(0_m, \sigma^2 I_m) \end{aligned}$$

于是我们有 $\text{ESS} = \eta^T \eta \stackrel{H_0}{\sim} \sigma^2 \chi_{(m)}^2$

现在我们可以构造线性约束检验问题 $H_0 : C\beta = h \leftrightarrow H_1 : C\beta \neq h$ 的检验统计量了:

$$\begin{aligned} F &:= \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/m}{\text{SSE}_{\text{full}}/n - p - 1} \\ &= \frac{\text{ESS}/m}{\text{SSE}_{\text{full}}/n - p - 1} \\ &\quad (\text{note that } \begin{cases} \text{ESS} \stackrel{H_0}{\sim} \sigma^2 \chi_{(m)}^2 \\ \text{SSE}_{\text{full}} \sim \sigma^2 \chi_{(n-p-1)}^2 \end{cases} \text{ where } H_0 : C\beta = h) \\ &\stackrel{H_0}{\sim} \frac{\sigma^2 \chi_{(m)}^2 / m}{\sigma^2 \chi_{(n-p-1)}^2 / (n - p - 1)} \\ &= F_{m, n-p-1} \end{aligned}$$

其中分子 $\frac{1}{m} \text{ESS} = \frac{1}{m} \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2$ 与分母 $s_{\text{full}}^2 = \frac{1}{n-p-1} \|y - X\hat{\beta}_{\text{full}}\|_2^2$ 的独立性由 $\hat{\beta}_{\text{full}} \perp s_{\text{full}}^2$ 保证.
(其中 s_{full}^2 记为全模型中根据 σ^2 的极大似然估计量构造出的无偏估计量)

我们记 $F_{m, n-p-1, \alpha}$ 为 $F_{m, n-p-1}$ 分布的 $1 - \alpha$ 分位数.

则线性约束 $C\beta = h$ 的显著性检验的 F -检验法为:

- (F -检验法)

$$\text{若 } F = \frac{\text{ESS}/m}{\text{SSE}_{\text{full}}/(n-p-1)} = \frac{\|A(C\hat{\beta}_{\text{full}} - h)\|_2^2/m}{\|y - X\hat{\beta}_{\text{full}}\|_2^2/(n-p-1)} > F_{m, n-p-1, \alpha}$$

则我们拒绝零假设 $H_0: C\beta = h$, 即我们认为线性先验关系 $C\beta = h$ 不成立.

$$\text{其中 } \begin{cases} A = X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} \\ \hat{\beta}_{\text{full}} = (X^T X)^{-1} X^T y \end{cases}$$

Problem 6

Consider the signal-plus-noise model such that

$$Y = \mu + \varepsilon \text{ where } \begin{cases} \mu \in \mathbb{R}^n \\ \varepsilon \sim N(0_n, \sigma^2 I_n) \end{cases}$$

(1) Find the BLUE of μ and its MSE [10 Marks]

(2) Consider another shrinkage estimator $\hat{\mu} := [\hat{\mu}_1, \dots, \hat{\mu}_n]^T$ such that $\hat{\mu}_i = Y_i(1 - \frac{a}{\|Y\|^2})$ ($i = 1, \dots, n$)

Find $a \in \mathbb{R}$ such that the MSE of $\hat{\mu}$ is smaller than the MSE of the BLUE [10 Marks]

Hint: use Stein's lemma: If $X \sim N(\mu, \sigma^2)$ then $E[(X - \mu)h(X)] = \sigma^2 E[h'(X)]$ with $h(X)$ continuous.

Part (1)

Find the BLUE of μ and its MSE [10 Marks]

Solution:

考虑 μ 的线性估计量 $\hat{\mu} = C^T Y$

要满足无偏性 $E[\hat{\mu}] = E[C^T Y] = C^T E[Y] = C^T \mu = \mu$ ($\forall \mu \in \mathbb{R}^n$)

矩阵 $C \in \mathbb{R}^{n \times n}$ 必须满足 $C^T \mu = \mu$ ($\forall \mu \in \mathbb{R}^n$)

这意味着 C 只能是单位阵 I_n

因此 $\hat{\mu}_* = Y$ 作为 μ 唯一的线性无偏估计量, 一定是最佳线性无偏估计量 (BLUE)

下面计算 $\hat{\mu}_* = Y$ 的 MSE:

$$\begin{aligned} \text{MSE}(\hat{\mu}_*) &= E[\|\hat{\mu}_* - \mu\|^2] \\ &= E[\|Y - \mu\|_2^2] \\ &= E[\|\varepsilon\|^2] \\ &= E[\text{tr}(\varepsilon^T \varepsilon)] \\ &= E[\text{tr}(\varepsilon \varepsilon^T)] \\ &= \text{tr}(E[\varepsilon \varepsilon^T]) \\ &= \text{tr}(\text{Cov}(\varepsilon)) \\ &= \text{tr}(\sigma^2 I_n) \\ &= n\sigma^2 \end{aligned}$$

Part (2)

Consider another shrinkage estimator $\hat{\mu} := [\hat{\mu}_1, \dots, \hat{\mu}_n]^T$ such that $\hat{\mu}_i = Y_i(1 - \frac{a}{\|Y\|^2})$ ($i = 1, \dots, n$)

Find $a \in \mathbb{R}$ such that the MSE of $\hat{\mu}$ is smaller than the MSE of the BLUE [10 Marks]

Hint: use Stein's lemma: If $X \sim N(\mu, \sigma^2)$ then $E[(X - \mu)h(X)] = \sigma^2 E[h'(X)]$ with $h(X)$ continuous.

Solution:

考虑 $\hat{\mu} = (1 - \frac{a}{\|Y\|^2})Y$, 我们有:

$$\begin{aligned}
\text{MSE}(\hat{\mu}) - \text{MSE}(\hat{\mu}_\star) &= \text{E}[\|\hat{\mu} - \mu\|^2] - \text{E}[\|\hat{\mu}_\star - \mu\|^2] \\
&= \text{E} \left[\left\| \left(1 - \frac{a}{\|Y\|^2}\right)Y - \mu \right\|^2 \right] - \text{E}[\|Y - \mu\|^2] \\
&= \text{E} \left[\|Y - \mu\|^2 + a^2 \frac{\|Y\|^2}{\|Y\|^4} - 2a(Y - \mu)^\text{T} \frac{Y}{\|Y\|^2} \right] - \text{E}[\|Y - \mu\|^2] \\
&= a^2 \text{E} \left[\frac{1}{\|Y\|^2} \right] - 2a \text{E} \left[(Y - \mu)^\text{T} \frac{Y}{\|Y\|^2} \right] \\
&= a^2 \text{E} \left[\frac{1}{\|Y\|^2} \right] - 2a \sum_{i=1}^n \text{E} \left[(Y_i - \mu_i) \frac{Y_i}{\|Y\|^2} \right] \quad (\text{use Stein's lemma and note that } Y_i \sim N(\mu_i, \sigma^2)) \\
&= a^2 \text{E} \left[\frac{1}{\|Y\|^2} \right] - 2a \sum_{i=1}^n \sigma^2 \text{E} \left[\frac{\|Y\|^2 - 2Y_i^2}{\|Y\|^4} \right] \\
&= a^2 \text{E} \left[\frac{1}{\|Y\|^2} \right] - 2a\sigma^2 \text{E} \left[\frac{n\|Y\|^2 - 2\sum_{i=1}^n Y_i^2}{\|Y\|^4} \right] \\
&= a^2 \text{E} \left[\frac{1}{\|Y\|^2} \right] - 2a\sigma^2 \text{E} \left[\frac{n\|Y\|^2 - 2\|Y\|^2}{\|Y\|^4} \right] \\
&= a^2 \text{E} \left[\frac{1}{\|Y\|^2} \right] - 2a(n-2)\sigma^2 \text{E} \left[\frac{1}{\|Y\|^2} \right] \\
&= a(a - 2(n-2)\sigma^2) \text{E} \left[\frac{1}{\|Y\|^2} \right]
\end{aligned}$$

要令 $\text{MSE}(\hat{\mu}) - \text{MSE}(\hat{\mu}_\star) < 0$, 等价于令 $a(a - 2(n-2)\sigma^2) < 0$
解得 $0 < a < 2(n-2)\sigma^2$

The End